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Methanogenic archaea diversity in hyporheic sediments of a small lowland stream



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ABSTRACT

Abundance and diversity of methanogenic archaea were studied at five localities along a longitudinal profile of a Sitka stream (Czech Republic). Samples of hyporheic sediments were collected from two sediment depths (0–25 cm and 25–50 cm) by freeze-core method. Methanogen community was analyzed by fluorescence in situ hybridization (FISH), denaturing gradient gel electrophoresis (DGGE) and sequencing method. The proportion of methanogens to the DAPI-stained cells varied among all localities and depths with an average value 2.08×10^5 per g of dry sediment with the range from 0.37 to 4.96×10^5 cells per g of dry sediment. A total of 73 bands were detected at 19 different positions on the DGGE gel and the highest methanogen diversity was found at the downstream located sites. There was no relationship between methanogen diversity and sediment depth. Cluster analysis of DGGE image showed three main clusters consisting of localities that differed in the number and similarity of the DGGE bands. Sequencing analysis of representative DGGE bands revealed phylotypes affiliated with members belonging to the orders Methanosarcinales, Methanomicrobiales and Methanocellales. The knowledge about occurrence and diversity of methanogenic archaea in freshwater ecosystems are essential for methane dynamics in river sediments and can contribute to the understanding of global warming process.

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1. Introduction

A great deal of attention has been focused at understanding the role of biodiversity in the natural environments [1]. In stream and river ecology, the biodiversity pattern has been probably influenced by River Continuum Concept (RCC) [2], which modeled spatial structure and function in streams in terms of gradual changes from headwaters to mouth. This concept predicts that for many common species, genetic diversity increases from upstream to downstream with a peak in mid-order streams [3]. In contrast to species like fish [4,5] or invertebrates [6,7], studies of the microbial biodiversity pattern along the longitudinal stream profile are still lacking. Although there is no clear definition of what “species” means for

microbes [8], the short generation time of microbes permits rapid changes in population abundance and changes of microbial community structure are often the result of varying environmental conditions. In a short time, the habitat of microbial community may shift from oxygen supersaturation to anoxia and new microbial niches are created, filled, destroyed and refilled [1]. The variety of microbial niches could probably be a reason why the microorganisms are immensely diverse.

In aquatic ecosystems, microbial life is often dominated by benthic biofilms (matrix-enclosed communities including bacteria, archaea, algae, fungi, and protozoa), which are essential for the decomposition of organic matter, the uptake and transfer of material and the microbial foodweb [9,10]. These microbial biofilms are formed in rivers on submerged surfaces such as stones, roots, river bed sediments and also sediments beneath and beside streams – hyporheic sediments [9,11]. The hyporheic zone, defined as sediments hydrologically linked to the open stream channel, is a highly dynamic ecotone with active exchange of water, dissolved

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material and organisms between river and groundwater [12,13]. The seemingly well-oxygenated hyporheic zone contains anoxic and hypoxic pockets (“anaerobic microzones”), which create a variety of microniches, allowing coexistence of a great diversity of microbial types [14–17]. The most useful way to consider the diversity of microbial types is in terms of functional groups – organisms that perform a particular process, for example converting complex organic compounds to carbon dioxide and methane [17]. Methanogenesis is a common process in habitats such as rice paddies [18], soils [19], lake sediments [20], as well as in the gastrointestinal tract of animals [21] and extreme habitats such as hydrothermal vents [22] and permafrost soils [23,24]. However, the production of methane is not a scarce process even in stream ecosystems, since a relatively high production of methane has been measured in river sediments [25–28].

Methane (CH_4) is produced mainly by methanogenic archaea as a final product of anaerobic respiration and fermentation [29,30]. Methanogenic archaea are obligate anaerobes, but anoxia is not the only factor influencing the process of methanogenesis. Other factors, which could stimulate the production of methane are low redox potential, increasing temperature and especially accessibility of substrates – acetate and hydrogen [31,32]. The acetate is converted into CH_4 and CO_2 by acetoclastic methanogens while hydrogenotrophic methanogens convert CO_2 and H_2 to CH_4 [33]. Because methanogenic assemblages are in environmental habitats responsible for the most of methane production, it is very important to study methanogen density and diversity in environments [34]. There are several studies of methanogen abundance and diversity in lake and sea sediments, paddy fields, rice roots and peatlands [34], but studies of methanogen diversity in hyporheic sediments are lacking.

Since microbial diversity is difficult to estimate, it is necessary to use molecular markers such as 16S rRNA. There are a variety of techniques which could be used for an assessment of microbial diversity and composition in natural samples [35]. Fluorescence in situ hybridization (FISH) is a powerful technique which allows to detect and identify the community composition directly in environmental samples. FISH is based on the direct microscopic identification of single cells after hybridization with 16S rRNA targeted fluorescent dye-labelled oligonucleotide probes [36,37]. Denaturing gradient gel electrophoresis (DGGE) is used to determine the genetic diversity of microbial communities. The procedure is based on electrophoresis of PCR-amplified 16S rDNA fragments in polyacrylamide gels containing a linearly increasing gradient of denaturants. In DGGE, DNA fragments of the same length but with different base-pair sequences can be separated. Separation is based on the electrophoretic mobility of partially melted DNA molecules in a polyacrylamide gel and resulting into a band pattern [38,39].

The aim of this study was to reveal the methanogen diversity of hyporheic sediments along the longitudinal stream Sitka profile. We focused on the detection of methanogenic archaea richness (DGGE), identification (sequencing) and quantification (FISH) in two hyporheic sediment depths (0–25 cm and 25–50 cm sediment layers) at five localities alongside the stream, simultaneously with measuring of physicochemical parameters (dissolved methane and oxygen saturation, potential methane production, organic carbon content etc.). Sitka stream is a relatively short, some 35 km long second-order stream with a mostly natural meandering channel. Our previous measurements showed that with increasing distance from the spring channel the slope decreases while sedimentation of fine particles and accumulation of organic matter within sediments increase. Similarly, dissolved oxygen concentration in the interstitial water is highest in the upstream part of the stream, gradually decreases toward the stream mouth and sharply decreases with sediment depth. Based on these findings and the theory that

microbial diversity is changing from headwaters to stream mouth along with River Continuum Concept, we hypothesized that the methanogen diversity and abundance would increase from upstream to downstream of the Sitka stream. Furthermore, based on our previous methane concentration measurements we hypothesized that methanogens occurring deep in the hyporheic sediments would exhibit higher abundance and taxonomic richness compared to those from upper sediment layer.

2. Material and methods

2.1. Study site

The sampling sites are located on the Sitka stream, Czech Republic. The Sitka is an undisturbed, second-order, 35 km long lowland stream originating in the Hrubý Jeseník mountains at 650 m above sea level. The catchment area is 118.81 km², geology being composed mainly of Plio-Pleistocene clastic sediments of lake origin covered by quaternary sediments. River bed sediments are composed of gravels in the upper parts of the stream while the lower part is characterized by finer sediment. Mean annual discharge is 0.81 m³/s. For more detailed characteristics of the sampling sites see [28,40] and Table 1.

2.2. Collecting of sediment and water samples

Five localities (no. 1–5) alongside stream profile were chosen for sampling sediment and interstitial water samples. The locality no. 1 is situated in headwater of the stream, while locality no. 5 is the most downstream. The map showing the location of study sites 1–5 is available in our previous study [28].

Hyporheic sediment and interstitial water samples were taken during summer 2011. The sediment samples were collected with a freeze-core method using N_2 as a coolant [41]. At each locality three sediment cores were taken. After sampling, surface sediment layer (0–25 cm) and layer of 25–50 cm in depth were immediately separated from each core. The particular layer samples were then pooled. The samples were then stored at a low temperature (4–6 °C) during the transport to the laboratory. After thawing the wet sediment of each layer was sieved and only particles <1 mm were considered for the following measurements, since most of the biofilm microorganisms are associated with this fraction [42]. Six sediment subsamples from each locality (three from each sediment layer) were used for subsequent microbial analysis.

Samples for estimation of total cell numbers and, consequently, for FISH analysis were fixed with paraformaldehyde (2% final conc.). Samples for measurement of organic matter content and determination of methanogenic potential were used immediately and samples for DNA extraction were frozen (–20 °C) until used.

Interstitial water samples were collected using a set of mini-piezometers [43] from both sediment depths (0–25 cm and 25–50 cm) at each study site. Three randomly replicated interstitial water samples were taken for measurement of particular physicochemical parameters.

The values of selected physicochemical and microbial parameters are expressed as means $\pm$ SE.

2.3. Analysis of environmental parameters

Sediment organic matter content was determined by oven-drying at 105 °C and subsequent combustion at 550 °C for 5 h. Organic matter values were then converted to carbon equivalents and expressed as a percentage according to the protocol in [44]. Sediment grain size distribution was analyzed from air-dried sediment samples and descriptive parameters were computed

Table 1

Selected physico-chemical and microbial parameters of the hyporheic interstitial water and sediment in two sediment depths of Sitka stream localities.

Variable/Locality	1	2	3	4	5
Grain median size [mm]	10.47 ± 1.90	9.41 ± 3.50	13.39 ± 0.20	0.59 ± 0.35	6.40 ± 1.05
	7.56 ± 4.82	12.66 ± 0.25	12.10 ± 1.10	0.36 ± 0.12	3.08 ± 2.27
Organic carbon in sediment < 1 mm [%]	1.09 ± 0.01	0.90 ± 0.01	0.79 ± 0.03	0.76 ± 0.01	0.52 ± 0.01
	1.06 ± 0.05	0.93 ± 0.01	0.36 ± 0.04	0.87 ± 0.09	0.94 ± 0.04
Interstitial dissolved oxygen saturation [%]	85.65 ± 2.56	88.18 ± 0.71	86.30 ± 0.62	67.14 ± 9.90	76.40 ± 3.49
	79.18 ± 1.23	86.38 ± 1.45	77.73 ± 2.72	14.85 ± 0.19	30.13 ± 3.45
Ferrous iron [mg/l]	ND	ND	ND	6.98 ± 2.52	5.96 ± 2.16
			1.79 ± 0.10	12.93 ± 0.31	4.88 ± 1.63
Acetate [mmol/l]	0.16 ± 0.01	0.36 ± 0.08	0.62 ± 0.07	1.61 ± 0.40	ND
	0.24 ± 0.07	0.30 ± 0.08	ND	1.99 ± 0.05	0.39 ± 0.07
Interstitial CH ₄ concentration [μg/l]	2.31 ± 0.08	0.88 ± 0.07	2.39 ± 0.06	3056.33 ± 1139.39	29.91 ± 16.74
	2.39 ± 0.11	0.75 ± 0.03	16.08 ± 7.93	3170.57 ± 832.69	122.73 ± 10.95
Methanogenic potential [μg CH ₄ /g DW/h]	1127.96 ± 341.10	178.18 ± 57.63	515.72 ± 32.97	16 506.48 ± 1853.65	1818.03 ± 167.59
	221.07 ± 180.64	164.74 ± 31.28	71.23 ± 2.64	212.08 ± 60.08	129.68 ± 6.07
Total cell number [× 10 ⁵ cells/g DW]	8.48 ± 2.71	5.14 ± 2.42	18.80 ± 6.63	41.03 ± 12.64	5.71 ± 1.62
	3.26 ± 0.69	4.04 ± 1.84	22.02 ± 5.66	15.95 ± 3.29	8.03 ± 4.91
Archaea abundance [× 10 ⁵ cells/g DW]	0.76 ± 0.37	0.19 ± 0.04	2.32 ± 0.17	1.60 ± 0.45	1.44 ± 0.59
	0.77 ± 0.23	0.57 ± 0.23	1.44 ± 0.41	1.18 ± 0.44	1.57 ± 0.60
Methanogenabundance [× 10 ⁵ cells/g DW]	0.37 ± 0.11	1.06 ± 0.24	4.42 ± 2.11	2.96 ± 1.90	0.48 ± 0.09
	0.87 ± 0.27	1.02 ± 0.33	2.99 ± 1.29	4.95 ± 1.42	1.63 ± 0.92

The grey rows indicate the variables measured in upper sediment layer (0–25 cm).

The white rows indicate the variables measured in lower sediment layer (25–50 cm).

ND = not detected (under detection limit).

using the SeDi database [45]. The rate of potential methane production (methanogenic potential) was measured in sediment using the PMP method [46]. Approximately 100 g (wet mass) of sediment with acetate addition (100 mg C in the incubation flask) was incubated for 72 h at 20 °C in the dark. Subsamples were taken every 24 h. Gas production was calculated from the difference between final and initial concentration and expressed per unit dry weight of sediment per hour (μg CH₄/g DW/h).

Concentrations of dissolved methane in the interstitial water were measured directly using a CHROM 5 gas chromatograph. Gas concentration in water was calculated using Henry's law (protocol by [47,48]). Results were expressed in μg CH₄ per liter of water. Concentration of acetate was measured using capillary electrophoresis equipped with diode array detector HP 3D CE Agilent (Waldbron, Germany) and was expressed in mmol per liter of interstitial water. Dissolved ferrous iron (Fe²⁺) concentration in interstitial water was measured using absorption spectrophotometry (DR-2000, Hach-Lange, Germany) after reaction with 1,10-phenanthroline. The values of ferrous iron concentration were expressed in mg Fe²⁺ per liter of water. Interstitial dissolved oxygen saturation (percentage) was measured directly in the field with a portable Hanna HI 9828 multimeter (Fisher Scientific, USA).

2.4. Direct cell counts

For removal of cells from the sediment particles, approximately 10 ml of fixed subsamples were sonicated for 3 × 30 s at 15% power (sonotrode MS 73, Sonopuls HD 2200, Sonorex, Germany), followed by incubation for 5 h under mild agitation with 10 ml of detergent mixture (Triton X-100, 0.5%, v/v; 0.1 M tetrasodium pyrophosphate and distilled water) and density centrifugation (protocol in [49]). For density centrifugation, the non-ionic medium Nycodenz (1.31 g/ml, Axis- Shield, Oslo, Norway) was used (Rotofix 32A, Hettich, Germany). 5 ml of Nycodenz was placed underneath 10 ml of treated sample and centrifuged at 3000 g for 60 min. 10 ml of supernatant was then taken for subsequent analysis.

The supernatants were filtered onto 0.22 μm polycarbonate membrane filters (GTP, Millipore, France). Filters were then cut into sections and stained for 10 min with DAPI solution (80 μg/ml, w/v, Sigma, Germany) in a dark cold place [50]. After staining the samples were rinsed in distilled water and 80% ethanol, air-dried,

mounted onto glass slides and counted on an epifluorescence microscope (Olympus BX 60, 1000 × magnification, Olympus corporation, Japan) equipped with a camera (Olympus DP 12). A total of 400 microbial cells were counted on at least 20 fields in three replicates and expressed per g of dry sediment (g DW).

2.5. Fluorescence in situ hybridization

We used a Cy3-labeled 16S rRNA-targeted oligonucleotide probe MPB1 5'-CAT GCA CCW CCT CTC AGC-3' [51] specific for methanogens and probe ARCH915.

5'-GTGCTCCCCCGCAATTCCT-3' targeting the domain *Archaea* [52]. The cells were hybridized according to the protocol in [53]. The supernatants which were used also for direct counts were filtered onto polycarbonate membrane filters (0.22 μm GTP, Millipore, France). The filters were then cut into sections and placed on glass slides. For the hybridization mixtures, 2 μl of probe-working solution was added to 18 μl of hybridization buffer into a microtube. Hybridization mix was added to the samples and the slides with filter sections were incubated at 46 °C for 3 h. After incubation, the sections were transferred into preheated washing buffer (48 °C) and incubated for 20 min in a water bath at the same temperature. The filter sections were washed and air-dried. The DAPI staining procedure followed as previously described. Finally, the samples were mounted in a 4:1 mix of Citifluor and Vecta Shield. The abundance of methanogens and archaea to the total number of DAPI stained cells was then calculated.

2.6. Nucleic acid extraction and PCR reaction

Nucleic acids were extracted from 0.3 g of sieved sediment with a Power Soil DNA isolation kit (MoBio, Carlsbad, USA) according to the manufacturer's instructions. DNA fragments (~350 bp) were amplified by PCR using a primer pair specific for methanogenic archaea. Primer sequences for 16S rDNA fragments are as follows, 0357 F-GC 5'-CCC TAC GGG GCG CAG CAG-3' (GC clamp at 5'-end CGC CCG CCG GCG GCG GCG GCGGG GCG GCG GCA CCG GGG G) and 0691 R 5'-GGA TTA CAR GAT TTC AC-3' [54].

PCR amplification was performed in TC-XP thermal cycler (Bioer Technology, Hangzhou, China) in 50 μl reaction mixture within 0.2 ml thin walled microtubes. The reaction mixture contained 5 μl

of $10 \times$ PCR amplification buffer, 200 μ M of each dNTP, 0.8 μ M of each primer, 8 μ l of template DNA and 5.0 U of FastStart Taq DNA polymerase (Polymerase dNTPack, Roche, Mannheim, Germany).

The initial enzyme activation and DNA denaturation were performed for 6 min at 95 °C, followed by 35 cycles of 1 min at 95 °C, 1 min at 55 °C and 2 min at 69 °C and a final extension at 69 °C for 8 min (protocol in [54]). PCR products were visualised by electrophoresis in ethidium bromide stained, 1.5% (w/v) agarose gel.

2.7. Denaturing gradient gel electrophoresis (DGGE)

DGGE was performed with an INGENYphorU System (Ingeny, Netherlands). PCR products were loaded onto a 7% (w/v) polyacrylamide gel cast in $1 \times$ TAE (40 mM Tris, 20 mM acetic acid, 1 mM EDTA, pH 7.4). The polyacrylamide gels (acrylamide: bisacrylamide, 37.5:1) were made with denaturing gradients ranging from 45 to 60%. 100% denaturant contained 7 M urea and 40% formamide. Electrophoresis was initially run at 110 V for 10 min at 60 °C, and afterwards for 15 h at 85 V. After electrophoresis, the gels were stained for 60 min with SYBR Green I nucleic acid gel stain (1:10 000 dilution, Lonza, Rockland, USA). DGGE gel was then photographed under UV illuminator (Molecular Dynamics). Images were arranged by image analysis software (NIS Elements, Czech Republic) and analyzed by programme Gel2K (University of Bergen, Norway).

2.8. Excising the bands and sequencing

Selected DGGE bands were excised from gel using a sterile scalpel blade. The bands were placed into tubes with 0.1 g of glass beads and 40 μ l of molecular grade water. The tubes with acrylamide fragments were vortexed briefly and incubated overnight at 4 °C. The supernatants were then used for reamplification using the primer pair (0357F and 0691R) without a GC-clamp [53]. The PCR products were then loaded onto agarose gel. After electrophoresis, the DNA fragments were excised from agarose and cleaned up with an UltraClean Gel Spin DNA Extraction kit (MoBio, Carlsbad, USA) according to the manufacturer's instructions. The DNA samples were then sent for sequence analysis to Macrogen company (Seoul, Korea).

Raw sequences data were examined by the program DECIPHER for chimeras checking. All sequences were then analyzed by BLAST software to search for the sequence identity among other methanogen sequences available in the GenBank database.

GenBank accession numbers are as follows:

KJ714048, KJ714049, KJ714050, KJ714051, KJ714052, KJ714053, KJ714054, KJ714055, KJ714056, KJ714057, KJ714058.

2.9. Statistics

The analyses were performed using statistical software R version 2.6.0. and GLM (General linear models). Total cell numbers (TCNs), values of methanogens and archaea detected by FISH and comparison of detected bands among all localities and depths were analyzed by a Negative Binomial Model and Wilcoxon signed rank sum test. All tests were considered significant at probability level $p < 0.05$.

DGGE band pattern was analyzed using Gel2K software (Svein Norland, Department of Biology, University of Bergen, Norway). This programme determined the presence or absence of DGGE bands and transformed the results to a binary file. The dendrogram was then constructed from these binary files using a Jaccard distance algorithm. The tree construction was performed by CLUSTER programme packaged with a Gel2K software.

3. Results and discussion

3.1. Environmental parameters of the study sites

The median grain size decreases with downstream distance (Table 1). Except locality no. 2, the values of grain size decreased with sediment depth. Because microbial abundance in sediments has exhibited a strong inverse correlation with sediment grain size [55], we reasoned that localities with the lower median grain size could host higher microbial community. Organic carbon content in sediment varied among all localities and depths with no considerable differences between sediment samples.

Interstitial water measurements revealed relatively high dissolved oxygen saturation except localities no. 4 and no. 5. Nevertheless, all localities showed a clear trend of decreasing oxygen concentration with sediment depth (Table 1). The downstream localities were also characterized by higher concentration of interstitial dissolved ferrous iron (Table 1). The presence of ferrous iron in the hyporheic zone provides indirect evidence of the consumption of oxygen [56]. The values of acetate concentration in interstitial water were similar among most of the localities and depths except for locality no. 4, where the acetate concentration reached values four times higher than at other localities. The presence of substrates like acetate or hydrogen and anaerobic conditions are essential for occurrence of methanogenic archaea [32].

The values of mentioned environmental parameters indicate, that downstream localities could provide preferable conditions for methanogen life (lower dissolved oxygen saturation, higher concentration of the ferrous iron, higher concentration of acetate and lower grain median size) compared to those from upstream. Indeed, interstitial methane concentrations as well methanogenic potential reached higher values at the downstream localities, while maximum methane concentrations were recorded at locality no. 4 (Table 1).

3.2. Total cell numbers

The total cell numbers (DAPI-stained cells, TCNs) of hyporheic sediments < 1 mm varied between localities and depths with the mean of 13.24×10^5 cells per g of dry sediment and the range between 3.26×10^5 and 41.03×10^5 cells per g DW (Table 1). Among the localities, significantly higher average values for TCNs were detected in the middle part (localities no. 3 and 4) of the stream ($p < 0.05$). The higher TCNs at these localities might be influenced by nutrients (phosphate and nitrate), because higher values of these nutrients originating from upstream located waste water treatment plant were detected in stream water at localities no. 3 and 4 (data not shown).

3.3. Archaeal and methanogen abundance

The microbial community of archaea and methanogens (identified with both applied probes - MPB1 and ARCH915) were detected for all localities and both sediment layers. The abundance of ARCH915-hybridized cells ranged between 0.19×10^5 (upper layer, locality no. 2) to 2.32×10^5 cells (upper layer, locality no. 3) per g DW (Table 1) with a mean of 1.18×10^5 cells. There was no significant difference between sediment depths ($p > 0.05$). The abundance of methanogens was significantly different among the localities ($p < 0.05$) but there was no significantly higher number of methanogens in lower sediment layer compared to upper sediment layer ($p > 0.05$). The average methanogen abundance accounted for 2.08×10^5 cells with the range of 0.37×10^5 (upper layer, locality no. 1) to 4.96×10^5 cells (lower layer, locality no. 4) per g DW of

sediment (Table 1). Among the localities, the highest average methanogen abundance was detected at locality no. 4 and the lowest value was found at locality no. 1.

The similar methanogen abundance was presented in several rice fields in Japan, China and India [57] and in sediments of Tokyo Bay in Japan [58]. The higher values of methanogen abundance were detected in soil of constructed wetland, Japan [59]. A higher archaea abundance than that found in our work was demonstrated in studies of siberian tundra soil [23] and lake sediments of Switzerland [60] and Germany [61]. It seems that the lower abundance of archaea and methanogens are related to low total cell numbers in our samples.

In this study the values of methanogen abundance were higher than numbers of archaea in hyporheic sediments of Sitka stream. The lower number of ARCH915-hybridized cells could be influenced by an unspecific binding of archaeal probe to some members of the domain *Bacteria* even under stringent hybridization conditions [62].

3.4. Methanogenic diversity by denaturing gradient gel electrophoresis

Methanogenic archaea associated with hyporheic sediments were compared by DGGE analysis along the longitudinal stream profile and among two different sediment depths. As shown in the band pattern (Fig. 1), the presence of DGGE bands in all samples indicates that methanogenic archaea occur at all localities up to 50 cm of the sediment depth. Analysis of DGGE image revealed a total of 73 detectable bands at 19 different positions. No one DGGE band appeared in all sediment samples, while unique bands were observed for localities no. 1 (two bands), no. 2 (one band) and locality no. 4 (two bands; Fig. 3). In general, there was a significant difference between numbers of DGGE bands among all localities

studied ($p < 0.05$). The highest number of bands was detected at locality no. 4, while the lowest number of bands was showed for locality no. 1 (Fig. 2). The highest band richness at locality no. 4 indicates the highest methanogenic diversity at this locality among all localities studied.

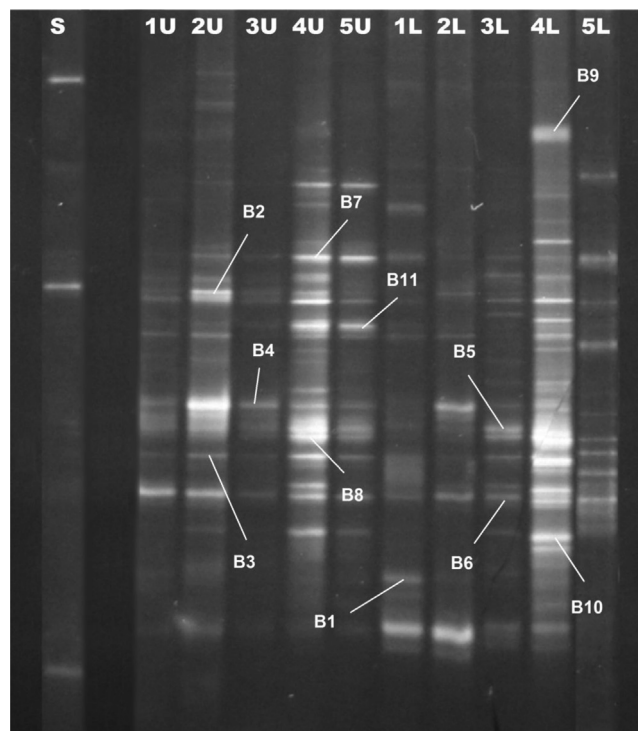
The number of total different bands (i.e. estimated diversity of methanogenic archaea) observed in this study was comparable with a number of the DGGE bands found in other studies. For example, Ikenaga and coworkers [63] found about 15–19 DGGE bands in their study of a methanogenic archaeal community in rice roots (Japan). Watanabe and colleagues showed 27 bands at different positions in the DGGE band pattern obtained from Japanese paddy field soil [64] and 20 different DGGE bands were detected in oxic soil microcosm of paddy field from Japan [65]. A relatively high number of archaeal bands (35 DGGE bands) was found in mycorrhizal samples of three different trees from a boreal forest, Finland [66].

Among the sediment depths, the number of DGGE bands ranged from 4 to 11 for the samples from upper sediment layer (0–25 cm) and from 5 to 13 for the samples from lower layer (25–50 cm), respectively. Generally, the number of DGGE bands varied with the sediment depths (Fig. 2) and we found no increasing number of DGGE bands with increasing depth ($p > 0.05$). The same results were showed in study of methanogen community in permafrost sediments of the Siberian Laptev Sea [67]. Zeng et al. [68] found no dependent relationship between bacterial diversity and sediment depth in Chinese eutrophic lakes. No relationship between sediment depth and methanogenic diversity is in contrast with study of Watanabe et al. [69] who found an archaeal community to be clustered according to two soil depths of paddy fields in Japan.

Our findings indicate that the methanogenic archaea diversity in Sitka stream is influenced rather by different location site than different sediment depth. In addition, it seems that locality no. 4 is the most preferable site for methanogenic archaea presence. We revealed not only optimal conditions for methanogen life and high methanogenic potential (Table 1) at this locality but also high methanogen abundance and diversity. These results are in concordance with our previous studies in Sitka stream [28,70].

3.5. DGGE band pattern along the longitudinal stream profile

Cluster analysis was performed for the DGGE pattern to evaluate methanogenic archaeal community in hyporheic sediments along the longitudinal profile of the Sitka stream. Cluster analysis revealed three main clusters consisting of localities that differ in the number and similarity of the DGGE bands (Fig. 3). The first cluster is composed of localities no. 4 (both sediment layers), no. 5 (both sediment layers) and no. 3 (lower sediment layer 25–50 cm) with



Description:
1–5: localities no. 1–no. 5;
U – upper sediment layer (0–25 cm), L – lower sediment layer (25–50 cm);
S – molecular standard;
excised bands are described as B1–B11

Fig. 1. DGGE band pattern obtained from hyporheic sediment samples along the longitudinal stream profile.

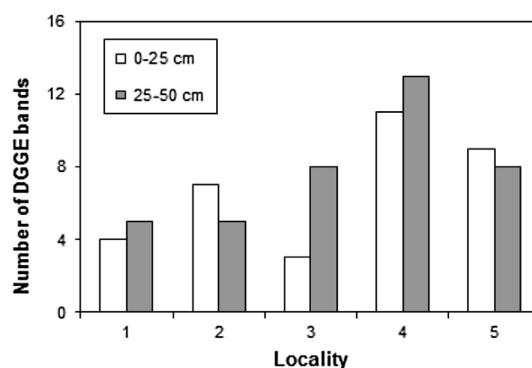


Fig. 2. Number of DGGE bands detected at studied localities and two different depths of hyporheic sediment.

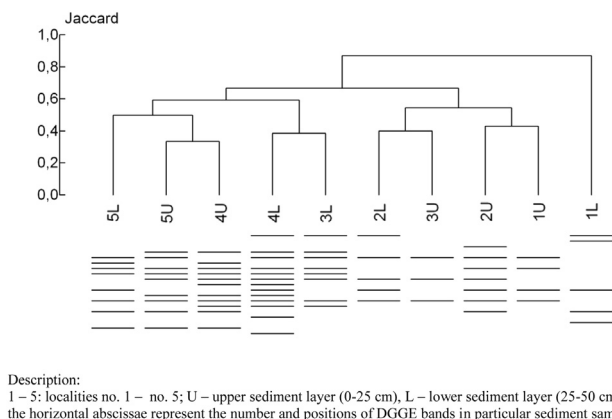


Fig. 3. Jaccard cluster diagram of DGGE bands from hyporheic sediments of the Sitka stream.

notably higher number of DGGE bands (8–13); the second cluster consists of localities of upper part of the stream – locality no. 1 (upper sediment layer 0–25 cm), locality no. 2 (both sediment layers) and locality no. 3 (upper sediment layer 0–25 cm); the third cluster is made of the lower sediment layer of locality no. 1 and stands separately. Both the second and third clusters showed less number of bands (3–7). Cluster analysis revealed also the highest Jaccard similarity for upper sediment layers of localities no. 4 and no. 5 among all samples studied.

The results of cluster analysis confirmed our hypothesis. We found the higher number of DGGE bands (i.e. higher methanogenic diversity) at sites located downstream (the first cluster) compared to localities being situated upstream (the second and the third cluster) ($p < 0.05$). These findings are in agreement with our prediction that methanogen diversity would increase from upstream to downstream. This hypothesis seems to be also supported with the environmental parameters measured at the downstream localities (Table 1), which offer the most suitable conditions for occurrence of methanogens. Generally, the expectation of higher downstream diversity is based on assumptions that downstream microbial biofilm communities recruit downstream dispersing propagules from upstream parts [71].

Some authors have shown a different trend of longitudinal changing in microbial diversity. For example McArthur et al. [72] studied the relationship between bacterial diversity and stream location and they discovered that genetic diversity of the studied species did not change longitudinally. Besemer and colleagues [3] showed the opposite findings than the results of our study. They determined that headwaters are critical reservoirs of microbial diversity for fluvial network. They explain their findings by intimate connection of headwaters with the terrestrial ecosystem and larger ratio of benthic surface area to water volume compared to

downstream. Nevertheless, these authors focused on bacterial diversity, while our study was aimed only on methanogenic archaea.

3.6. Sequencing of representative DGGE bands

A total of 13 excised bands from all localities were sequenced. The cut-off value for determination of identical sequences was 90%.

Eleven bands were successfully sequenced and their affiliation to the orders Methanosarcinales, Methanomicrobiales and Methanocellales was confirmed. Six band-sequences were affiliated to the family Methanosarcinaceae and the nearest identical uncultured sequences were obtained from an anaerobic digester in USA [73], sewage sludge of a methane fermentor in Japan [74], rice rhizospheres in Germany [75,76], a high-temperature petroleum reservoir in China [77] and anoxic rice roots in Italy [78]. The band-sequences affiliated to the family Methanosarcinaceae were isolated from all localities except locality no. 1 (Table 2). Three band-sequences were related to the acetoclastic methanogen *Methanosaeta* sp. and showed the highest similarity with sequences originated from anaerobic batch reactor in USA [79], swine manure anaerobic digester in China [80] and sediment sample of hydrocarbon-contaminated aquifer in Germany [81]. The *Methanosaeta* affiliated sequences were retrieved from localities no. 3 and no. 4 (Table 2). The Methanosarcinaceae members can utilize all methanogenic substrates except for formate while the members of family Methanosaetaceae grow exclusively using acetate as an energy source [82].

One single band was affiliated to the family Methanomicrobiaceae and the closest uncultured sequence was obtained from anoxic sediment of sub-saline shallow lake in Spain [83]. The Methanomicrobiaceae-like sequence was detected at locality no. 4 (Table 2). Methanogens belonging to the order Methanomicrobiales grow only in the presence of hydrogen, formate and alcohols with the exception of methanol [82].

Another single band was related to the family Methanocellaceae and the nearest uncultured sequence was retrieved from rice roots in Germany [76]. This unique band was detected at locality no. 1 (Table 2). *Methanocella* sp. is a novel hydrogenotrophic methanogen, which belongs to the order Methanocellales previously known as archaeal group Rice Cluster I from Japanese rice paddy field. The members of the order Methanocellales are known to utilize H_2 or formate for growth and methane production [84].

The twelfth sequence retrieved from lower sediment layer of locality no. 5 was affiliated to family Methanosarcinaceae, but the sequence was only 301 bp long and could not confirm accurate affiliation to methanogenic archaea. The last sequence, which originated from lower sediment layer of locality no. 4, was found to be a possible chimera composed probably of sequences from *Methanosphaerula* sp. *Methanosphaerula* is a novel genus belonging to the order Methanomicrobiales. This recently described taxon

Table 2
Identity of representative DGGE band-sequences detected by PCR-DGGE in hyporheic sediments of Sitka stream.

Phylogenetic affiliation	DGGE band	Locality	GenBank accession no.	Closest sequence – origin and accession no.	Sequence identity (%)
Methanocellaceae	B1	no. 1	KJ714049	Rice roots, Germany (AJ699080)	93
Methanosarcinaceae	B2	no. 2	KJ714050	Anaerobic digester, USA (JN052760)	99
	B3	no. 2	KJ714051	Sewage sludge fermentor, Japan (AB721088)	98
	B5	no. 3	KJ714048	Rice rhizosphere, Germany (AJ879040)	90
	B6	no. 3	KJ714057	Anoxic rice roots, Italy (AM050406)	99
Methanosaetaceae	B7	no. 4	KJ714054	Rice roots, Germany (AJ699125)	99
	B11	no. 5	KJ714056	Petroleum reservoir, China (JN882370)	98
	B4	no. 3	KJ714052	Anaerobic reactor, USA (JQ684620.1)	98
	B8	no. 4	KJ714055	Anaerobic digester, China (KC582669)	97
Methanomicrobiaceae	B10	no. 4	KJ714058	Hydrocarbon-contaminated aquifer, Germany (JQ087853)	99
	B9	no. 4	KJ714053	Anoxic lake sediment, Spain (HQ003660)	99

inhabits peat forming wetlands and grows only with hydrogen and formate [85]. Although the sequence from locality no. 4 had a 95% match to database sequence of *Methanosphaerula* sp., this band may be an artifact produced by the PCR analyses.

Our results indicate the presence of both hydrogenotrophic and acetoclastic methanogens in hyporheic sediments along the longitudinal profile of Sitka stream. These findings confirmed the results of our previous study on stable carbon isotope ($\delta^{13}\text{C}_{\text{CH}_4}$) measurement [70], which revealed that both acetoclastic and hydrogenotrophic pathways take part in the methanogenesis process in the Sitka stream. The detection of methanogens belonging to the orders Methanosarcinales and Methanomicrobiales, in the present study is supported by Buriánková et al. [86] who studied the phylogenetic composition of methanogens at locality no. 4 using cloning techniques and revealed 26 different *mcrA* gene sequences affiliated with the orders Methanomicrobiales, Methanosarcinales and Methanobacteriales. Chaudhary et al. [87] also detected 74 different 16S rRNA gene sequences related to the order Methanosarcinales. The differences between these three studies are in detection of methanogens belonging to the orders Methanocellales (this study) and Methanobacteriales [86].

The results of our study are also comparable with studies from environments such as paddy fields, anaerobic reactors and rice rhizospheres. Ikenaga et al. [63] revealed the members of the orders Methanomicrobiales and Methanocellales in rice roots in Japan, and Hwang and coworkers [88] studied methanogenic archaeal communities of anaerobic sludge reactor (Japan) and revealed sequences were related to the orders Methanosarcinales, Methanomicrobiales and Methanobacteriales. Moreover, Watanabe et al. [64,69] detected paddy field soil sequences affiliated with the orders Methanomicrobiales, Methanocellales, Methanosarcinales and Methanobacteriales.

4. Conclusions

We confirmed the presence of methanogenic archaea along the longitudinal profile of the Sitka stream up to 50 cm of sediment depth, and that methanogenic diversity increased from upstream to downstream. Downstream located sites characterized by accumulation of finer sediment grains, lower dissolved oxygen concentration and higher acetate concentration create the most preferable sites for methanogens and their activity. However, we did not confirm the influence of sediment depth on methanogen diversity and abundance. Sequencing analysis revealed the occurrence of both hydrogenotrophic and acetoclastic methanogens affiliated with the orders Methanosarcinales, Methanomicrobiales and Methanocellales. We also confirmed that the combination of FISH, DGGE and sequencing methods can provide us valuable information about methanogenic abundance, composition and diversity which may be crucial for an understanding of the methane cycle in stream ecosystems.

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